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Kurilla-Pokrop studio
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AD 2

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Studio Brief

the studio asks us to rethink logistics architecture as we know it today. Through a research-based design approach, the studio examines how contemporary logistics systems function, how they are spatially organized, and how their typologies shape urban and ecological environments.

The first phase of the studio focused on research into existing logistics models, including site visits to active logistics facilities and a series of expert lectures from professionals working in development, architecture, landscaping, and economics. These engagements provided valuable insight into the operational realities, environmental performance, and spatial limitations of current logistics architecture.

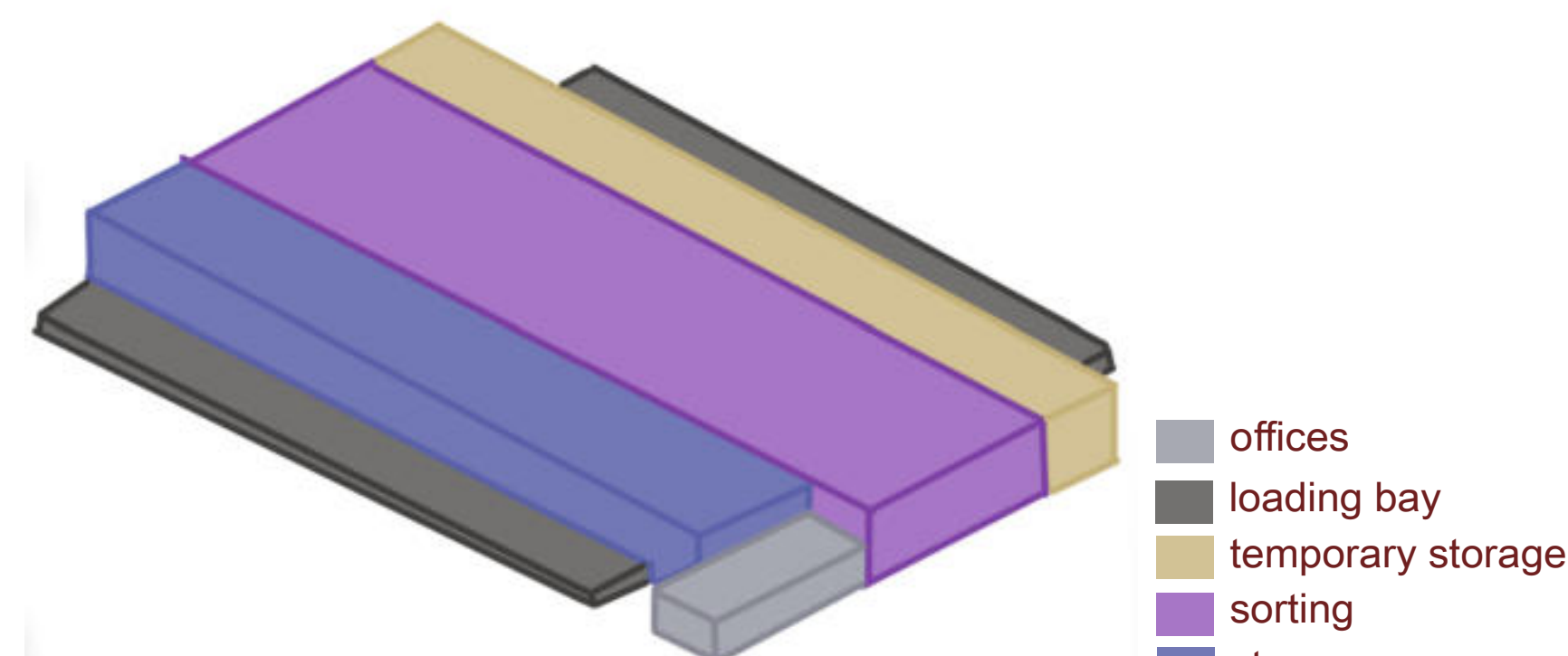
Building on this research, we were challenged to find a niche within the broader logistics landscape, conduct case studies of existing facilities in that niche and propose original design innovations that respond to emerging challenges we analysed.

The studio emphasized the creation of innovative designs based on tangible evidence backed by measurable performance criteria and key performance indicators relevant to our individual projects.

The studio ultimately positions logistics architecture as an active participant in shaping resilient urban futures, exploring how these typically isolated infrastructures can be redesigned to benefit its different stakeholders.

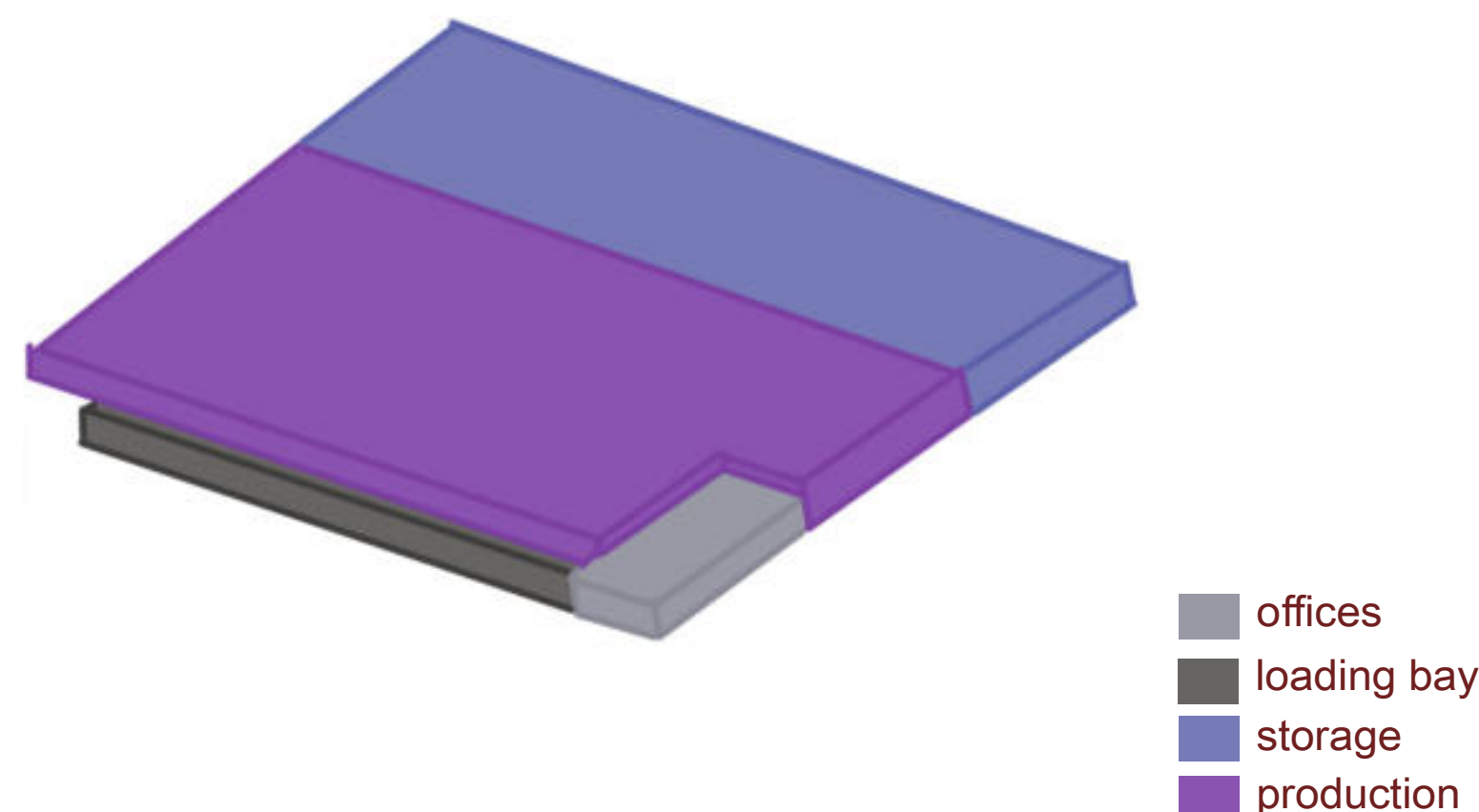
Types of Logistics Buildings

cross dock



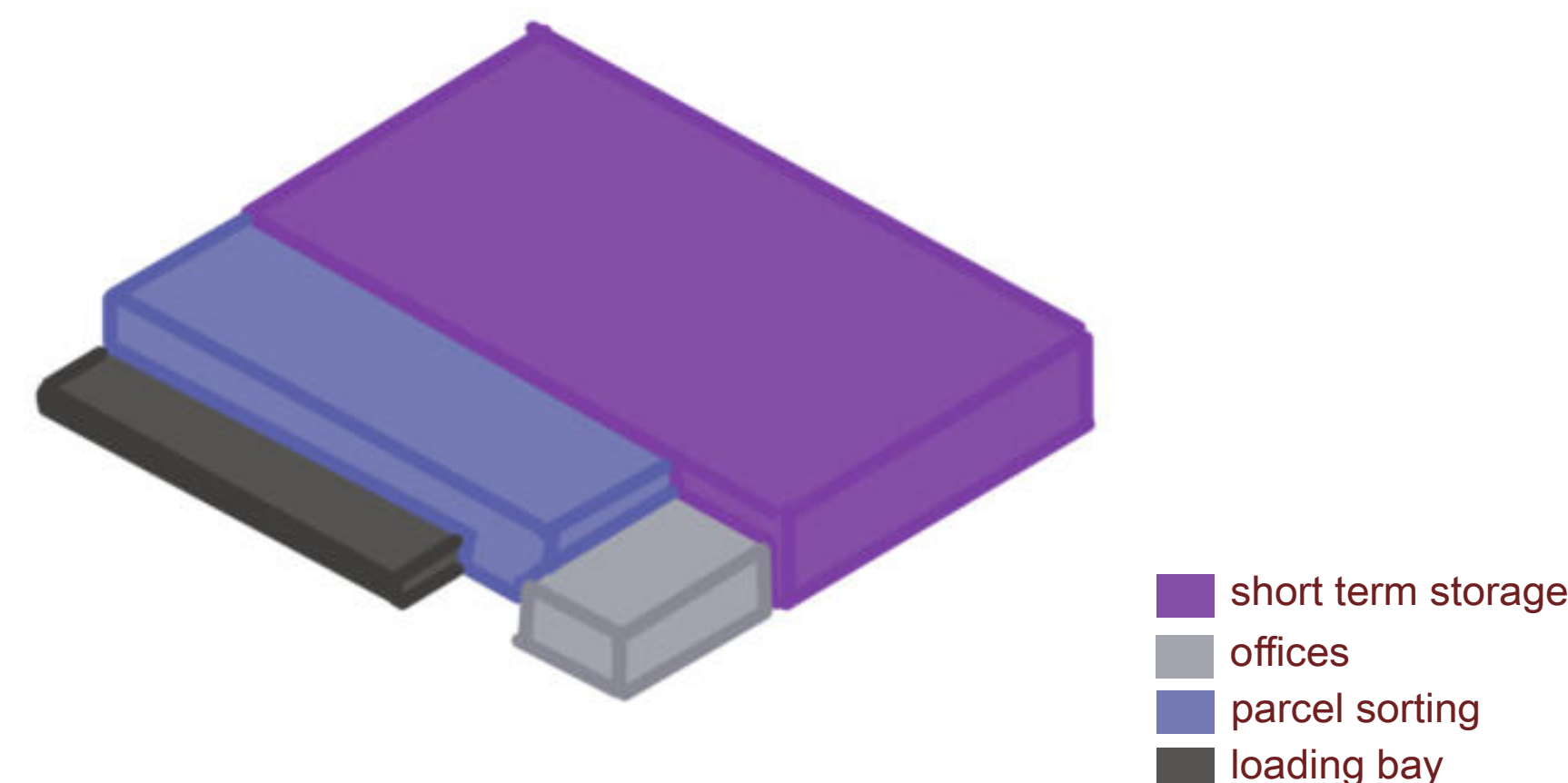
A logistics building designed for the rapid transfer of goods between incoming and outgoing trucks with minimal storage time. Its primary purpose is speed and operational efficiency. These facilities are generally shallow in depth, with loading docks on opposite sides, and are often located near major road networks or freight corridors.

light industrial



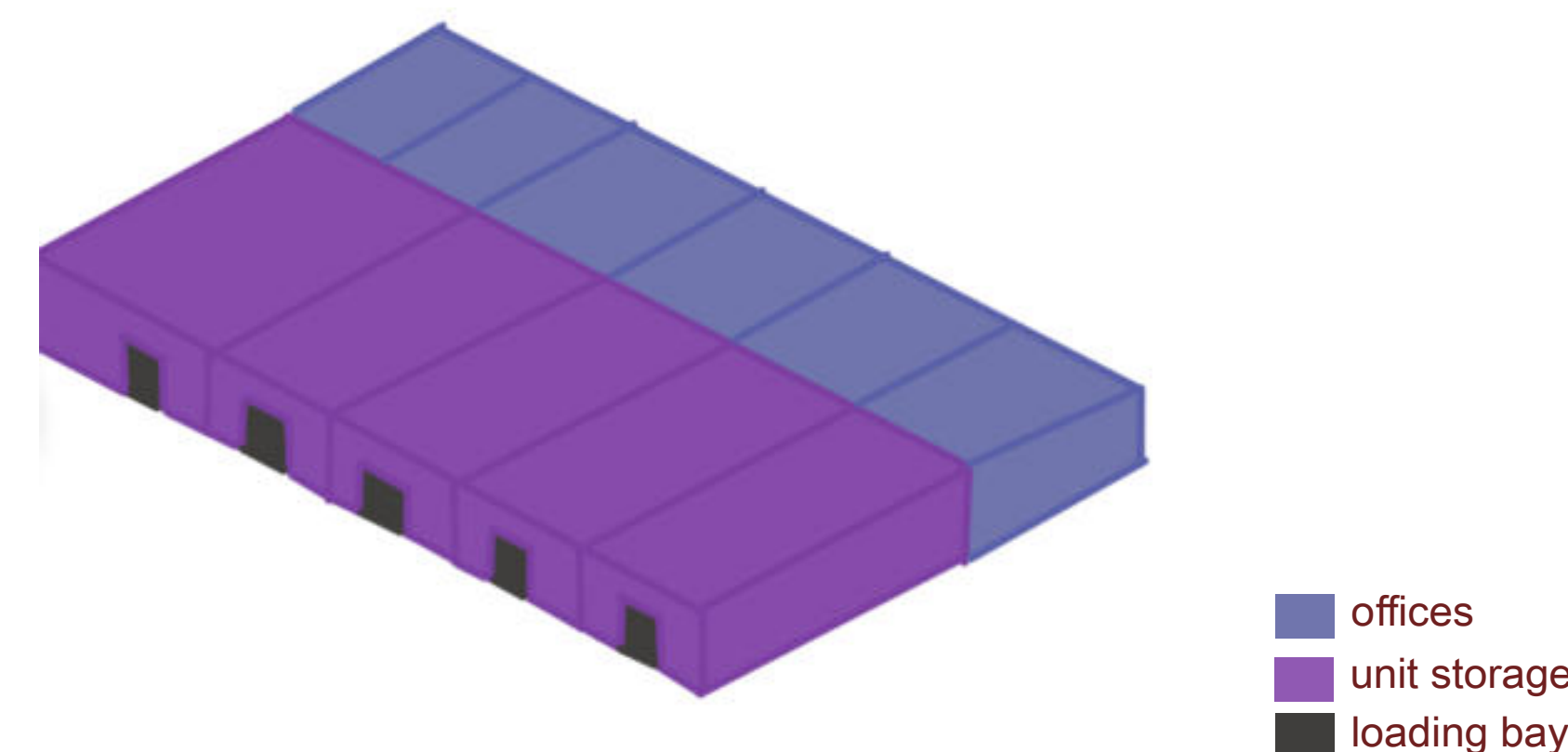
A hybrid logistics typology combining warehousing with assembly, packaging, repair, or specialized technical production. It supports light manufacturing processes alongside storage and distribution. These buildings often include office space and are commonly found in mixed industrial-commercial districts.

last mile delivery



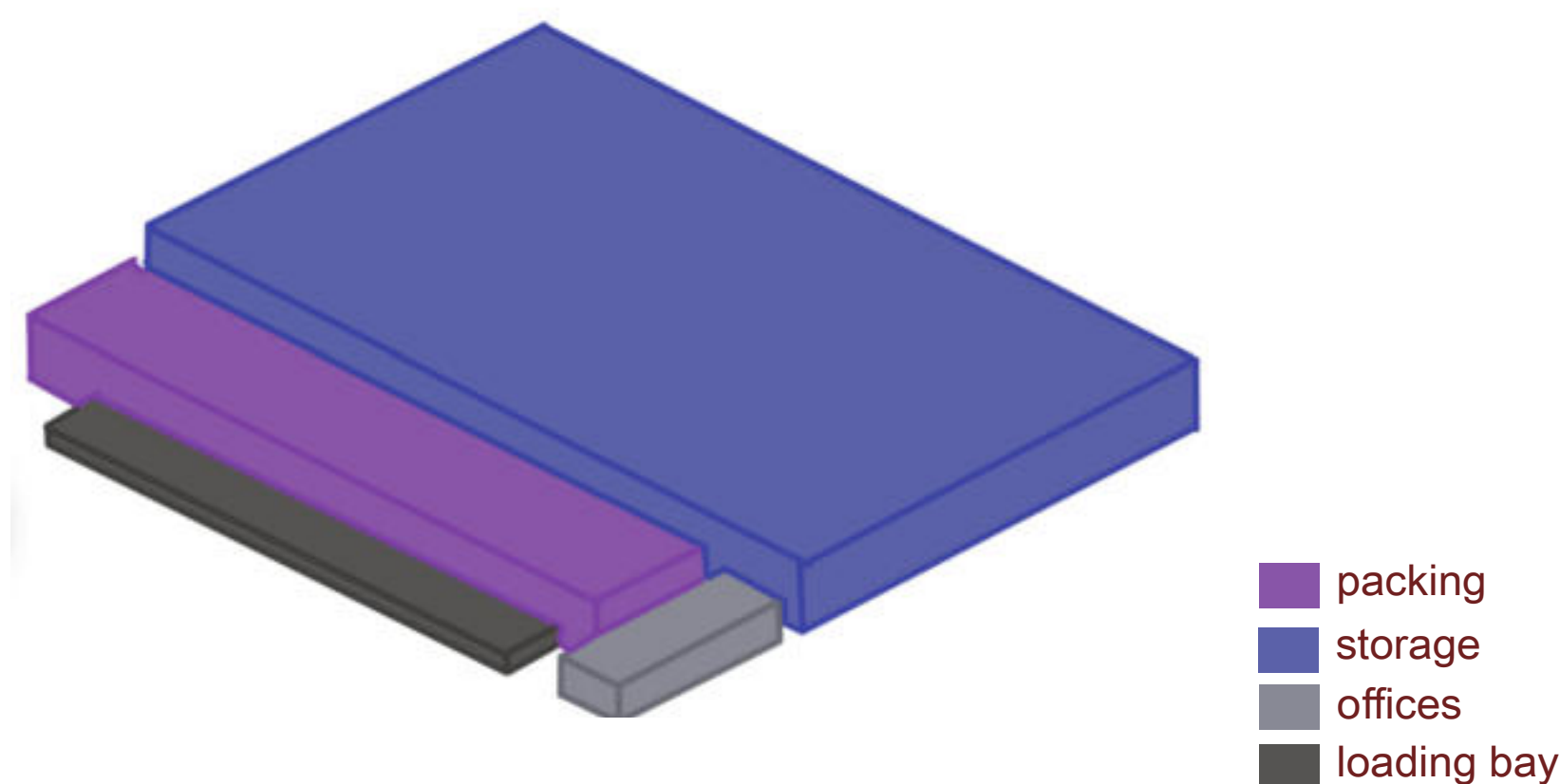
A smaller logistics node positioned close to urban areas for final package distribution to customers. It supports fast local delivery and short-term parcel handling. These facilities are usually compact, highly accessible, and often located at the edge of dense urban neighborhoods.

multi tenant



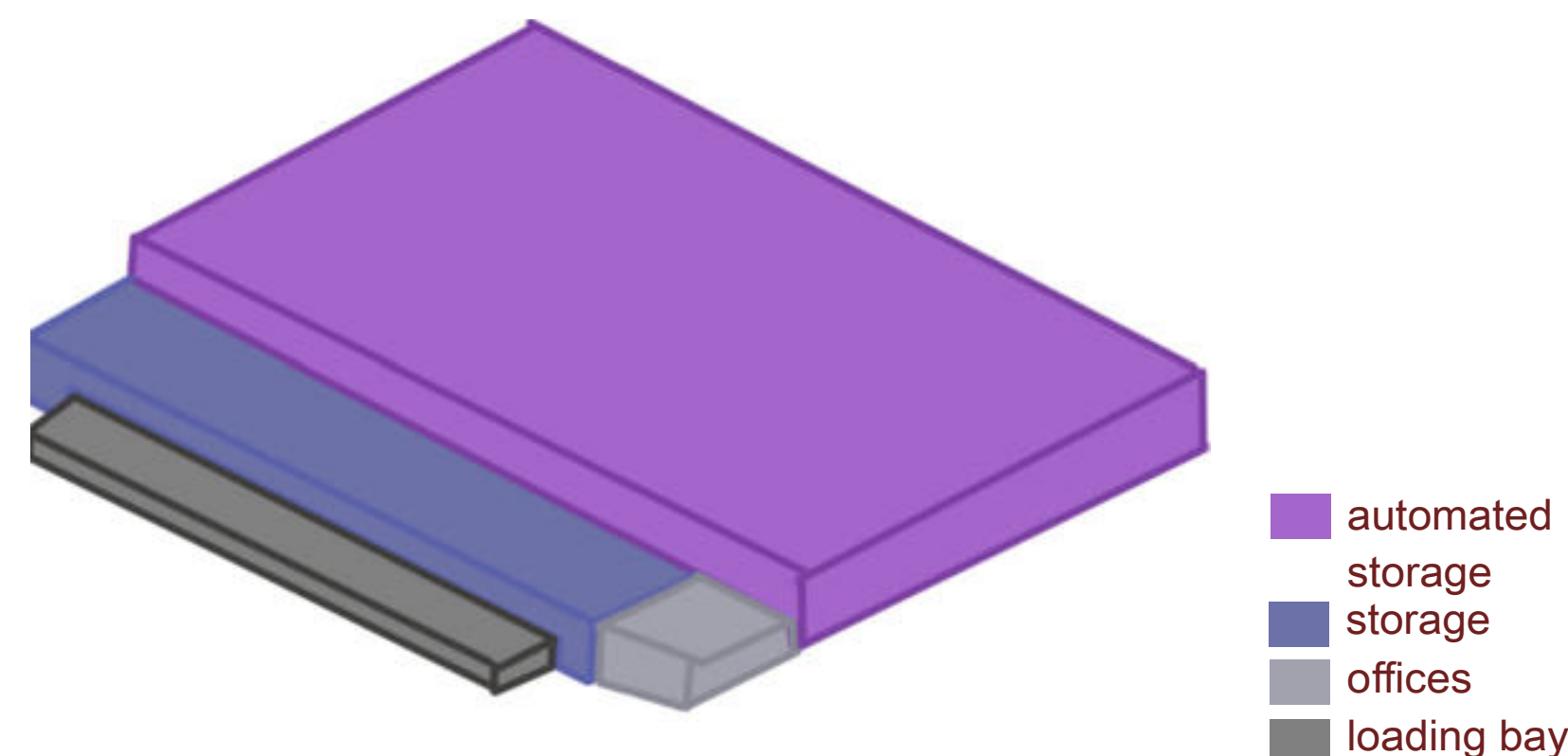
A flexible logistics building divided into multiple leasable units for different operators. It allows shared infrastructure while accommodating varied business needs. These facilities are modular in design, adaptable in layout, and commonly located within industrial parks or commercial logistics zones.

big box



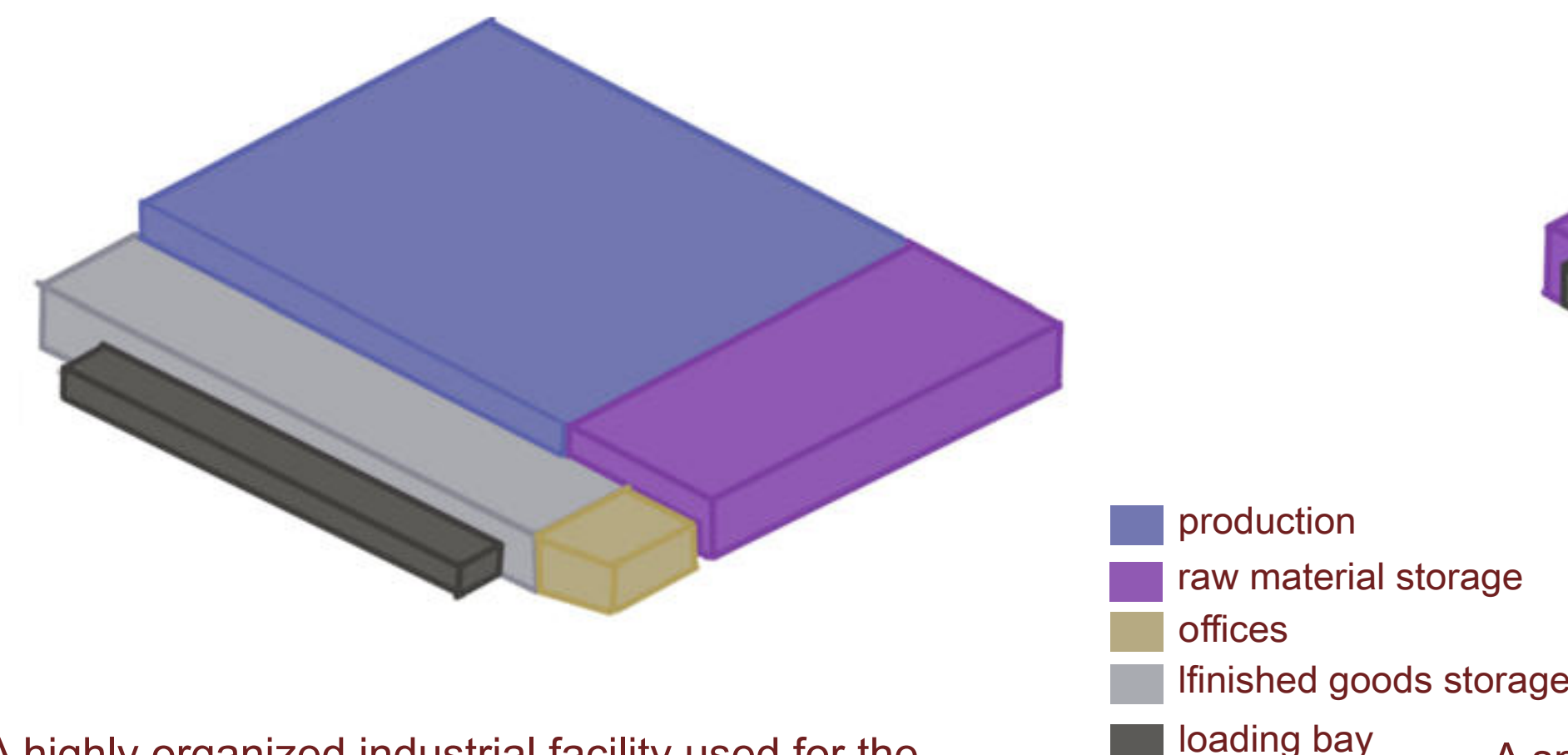
A large-scale logistics warehouse used for the storage and regional distribution of goods. These facilities handle bulk inventory and support long-distance supply chains. They are typically single-story, highly standardized buildings with large truck courts and are usually located on urban peripheries near highways or major transport infrastructure.

automated warehouse



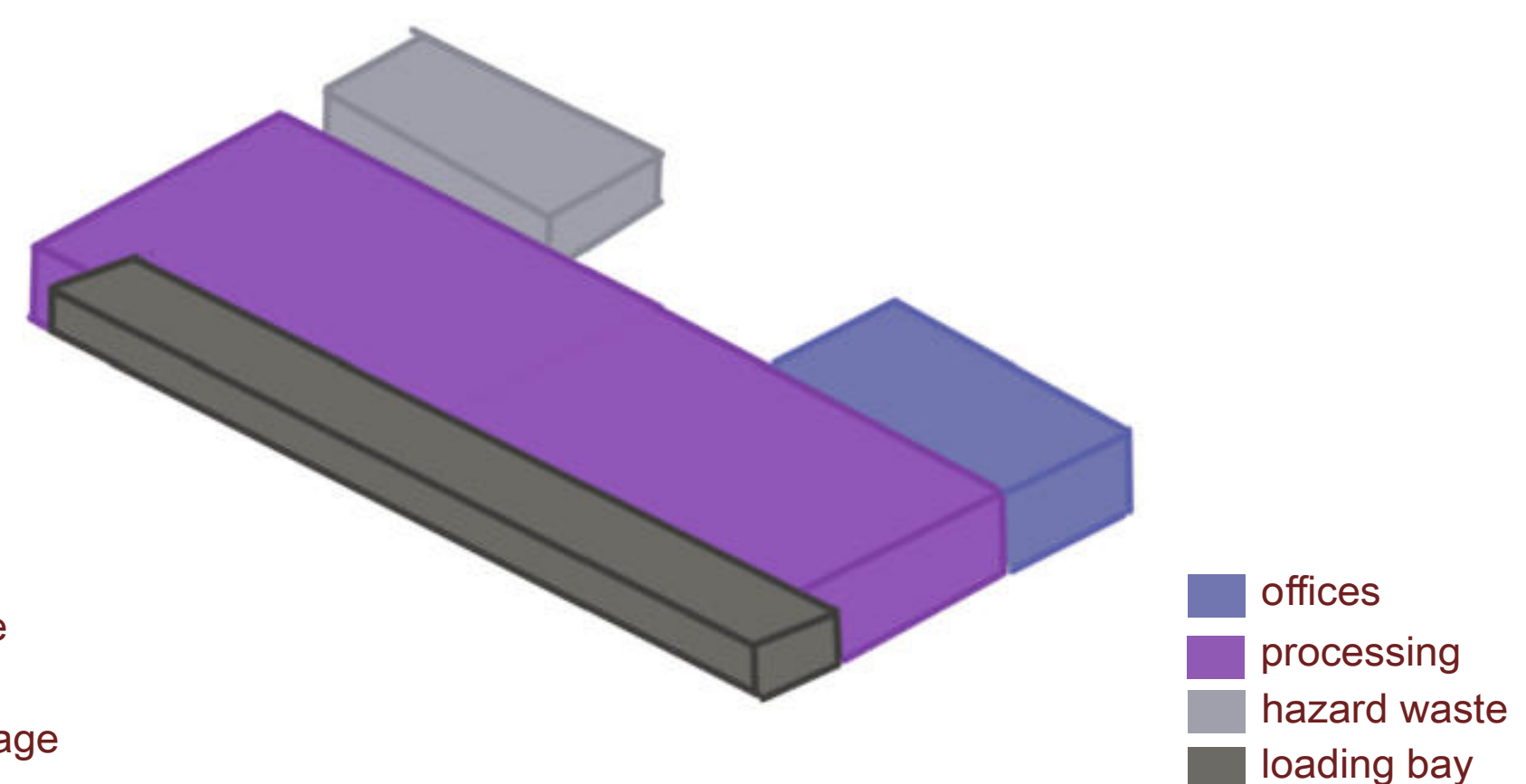
An automated warehouse is a highly mechanized logistics facility designed for the storage, retrieval, and movement of goods using robotic systems and automated handling technologies. These buildings often rely on vertical storage systems, conveyor networks, and minimal manual labor to maximize operational efficiency. They are typically compact, high-density facilities located near major transport infrastructure to enable rapid and scalable distribution.

Industrial manufacturing



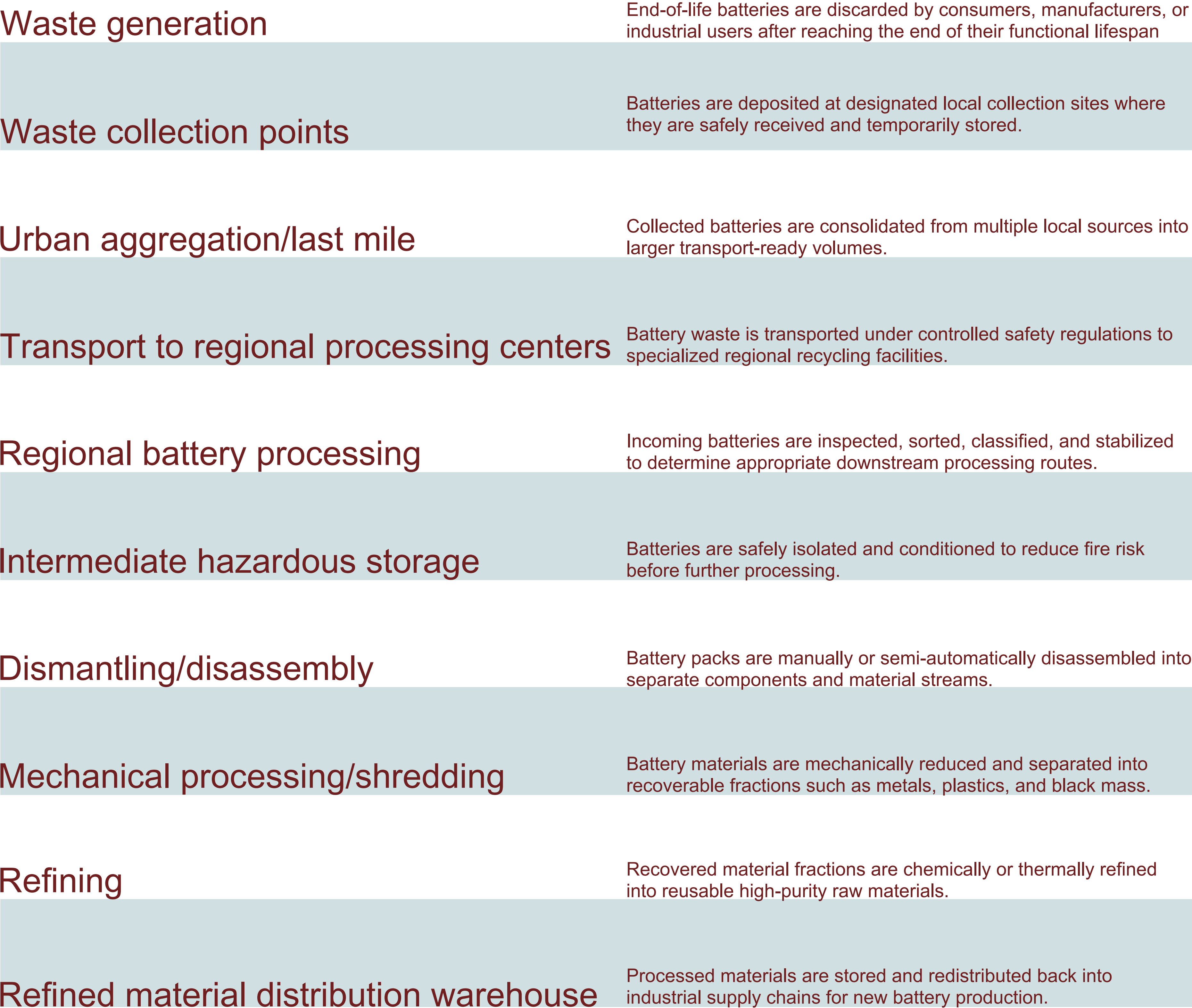
A highly organized industrial facility used for the production and assembly of goods through machining, fabrication, and assembly processes. These buildings often include specialized equipment, production lines, and integrated quality control systems. They are typically large-scale facilities located in industrial zones with strong access to transport infrastructure and supply chains.

recycling

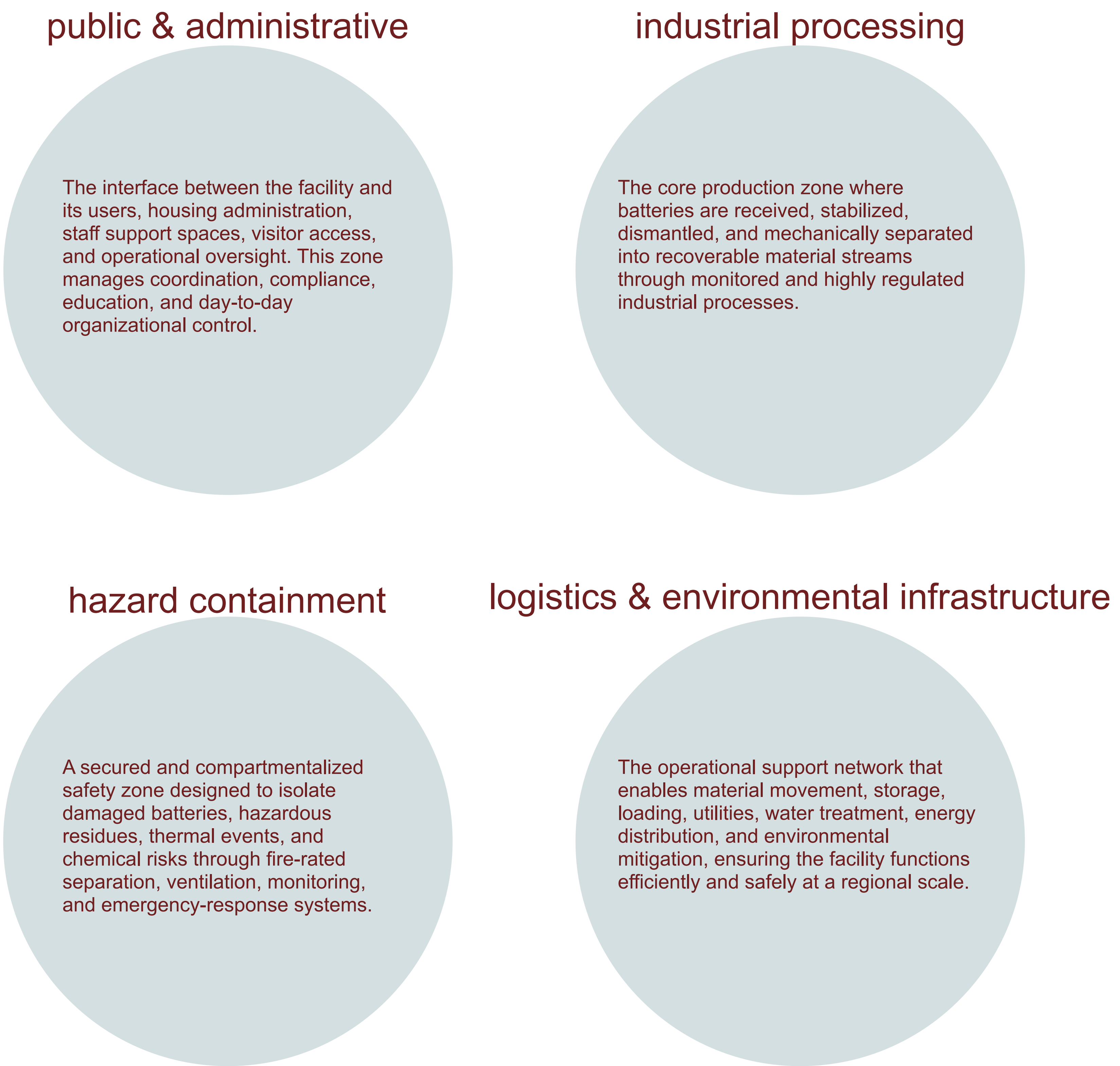


A specialized logistics facility used for sorting, stabilizing, dismantling, and preparing recyclable or hazardous materials for further processing. It combines logistics with controlled industrial treatment systems and often requires strict safety separation, ventilation, and environmental controls. These facilities are typically located in industrial zones with controlled access and infrastructure capacity.

Batteries Recycling Process



Functions of a Regional Processing Facility



case study 01

REDUX Recycling Facility

Location: Bremerhaven, Germany

Operator: REDUX Recycling

Opened: 2018

Processing capacity: 10,000 tonnes annually

5 tonnes/hour

Facility type:

Industrial lithium-ion battery pre-processing and mechanical recovery plant

Site context:

Heavy industrial-port logistics zone

Urban condition:

Highly separated from residential context

- Main Characteristics**
- monolithic industrial massing
 - process-driven spatial organization
 - truck-access optimized circulation
 - minimal landscape integration

- Key Takeaways**
- Strengths:
- strong hazardous process segregation
 - clear operational sequencing
 - industrial-scale technical credibility
- Limitations:
- no contextual softening
 - no public interface
 - little ecological integration
 - visually hostile industrial expression



Located within Bremerhaven’s industrial-port logistics district, the REDUX facility represents one of Europe’s first large-scale industrial lithium-ion battery recycling plants. Designed as a highly specialized processing environment, the plant receives batteries from across regional and national collection networks before discharging, dismantling, thermally pre-treating, and mechanically separating recoverable material streams. Its location within an established industrial landscape allows operational efficiency and logistical accessibility, but reflects the conventional industrial model of spatial isolation and minimal public or ecological integration. The facility demonstrates high technical performance and process specialization, while highlighting the limited architectural and environmental responsiveness typical of contemporary recycling infrastructure.

case study 02

Li-Cycle Germany Spoke

Location: Sülzetal (near Magdeburg), Saxony-Anhalt

Operator: Li-Cycle

Opened: 2023

Facility type: Regional modular battery recycling “Spoke” pre-processing facility

Operational role: Regional intake, shredding, black mass production for downstream refining

Site context: Industrial development zone / greenfield industrial cluster

Strategic location: Close to major battery manufacturing ecosystem in central Germany

- Main Characteristics**
- modular logistics massing
 - optimized regional accessibility
 - Adaptable process capacity

- Key takeaways**
- strengths:
- distributed network efficiency
 - modular adaptability
 - scalable process architecture
- Limitations:
- weak site-specific identity
 - limited ecological ambition
 - minimal human-scale response
 - generic industrial typology



The Li-Cycle Germany Spoke in Sülzetal operates as a decentralized regional processing node within Li-Cycle’s distributed recycling network model. Strategically positioned near major battery manufacturing and automotive supply-chain infrastructure, the facility receives battery waste and production scrap for safe discharge, mechanical processing, and black-mass generation before forwarding refined output to larger downstream recovery hubs. Unlike traditional centralized recycling plants, the modular “Spoke” typology prioritizes logistical flexibility, scalability, and proximity to production ecosystems. Architecturally, however, the facility remains representative of contemporary logistics development: operationally efficient but formally generic, with limited integration into landscape or surrounding urban context.

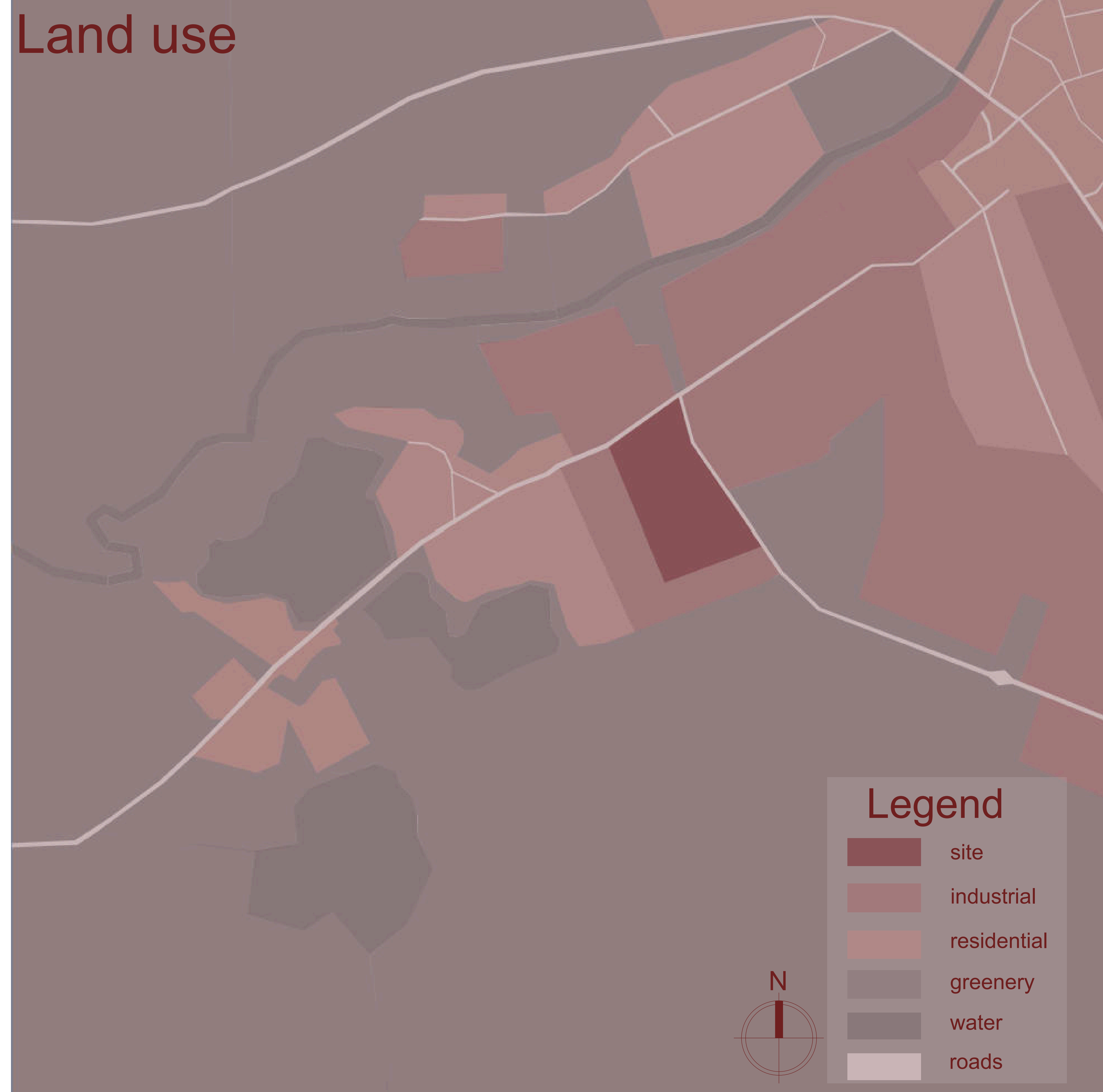
Project Context

Located on the southern urban fringe of Česká Lípa, the pragometal recycling facility is located between light-industrial infrastructure, dispersed residential development, and green landscape. The facility operates as a regional collection and pre-processing node for secondary metal and battery waste.

Pragometal benefits from near to ideal logistical accessibility, but simultaneously reflects the spatial characteristics of conventional industrial development: asphalt-dominated external yards, exposed technical surfaces, and minimal ecological or social integration with its immediate surroundings. Positioned in a mixed industrial-residential edge zone, the site produces a direct spatial tension between heavy logistical function and the adjacent human-scale development.

This condition reveals a broader architectural problem within logistics infrastructure, where performance and operational efficiency are prioritized, often at the expense of environmental quality, public perception, and long-term urban compatibility. As circular economies expand and battery waste management becomes increasingly critical, it is imperative to rethink this kind of architecture, and reinvent it in a way that will allow it to balance technical performance with ecological and social responsibility.

My project proposes a complete remodeling of the recycling facility, using only the original location and site boundaries. I will design a pre-processing/light processing recycling facility that focusses only on battery waste.



Users & Needs

workers

- daylight access
- clean air/ventilation
- safe working conditions
- efficient circulation
- break spaces

truck drivers

- efficient site access
- fast loading/unloading
- safe maneuvering spaces
- break spaces

residents

- visual buffer
- reduced industrial noise
- minimized truck disturbance
- attractive streetscape
- safety measures

tourists

- safety measures
- pleasant visual environment
- non-hostile industrial interface

local wildlife

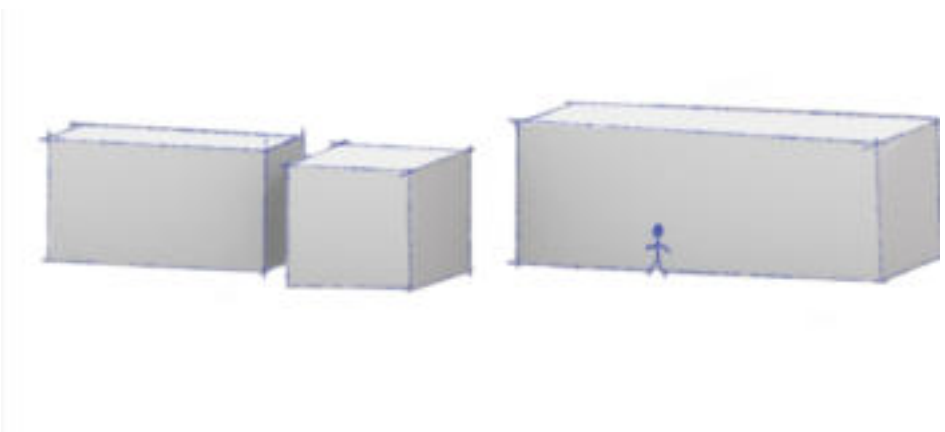
- habitat continuity
- native plants
- reduced habitat fragmentation
- water retention capacity
- nesting & shelter
- bio corridors

emergency/fire services

- hazard isolation
- rapid access routes
- direct access to containment zones
- visual site legibility

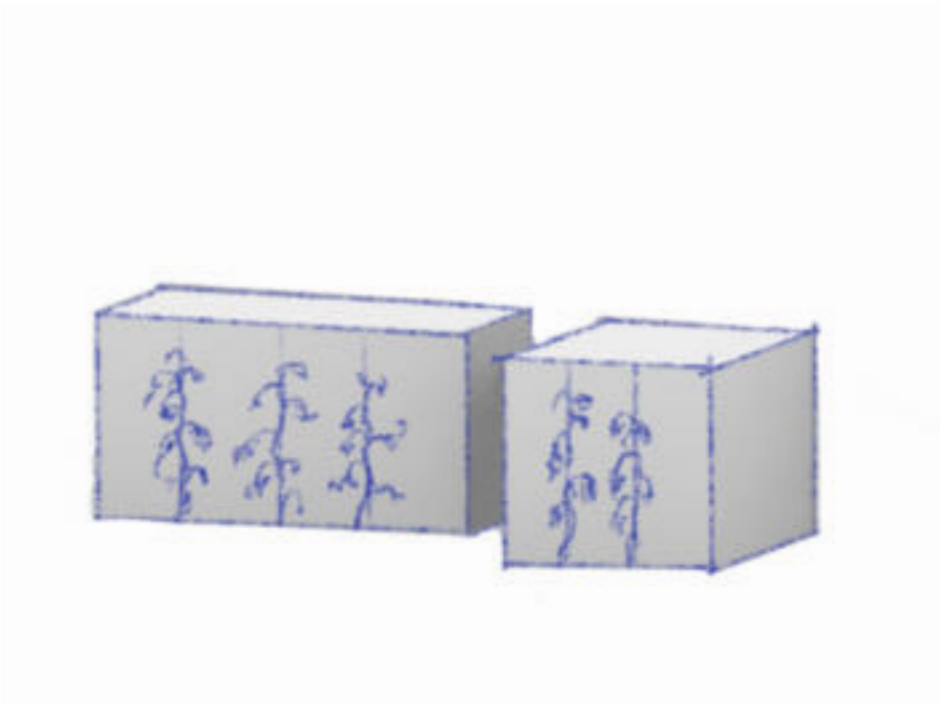
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Concept Proposal



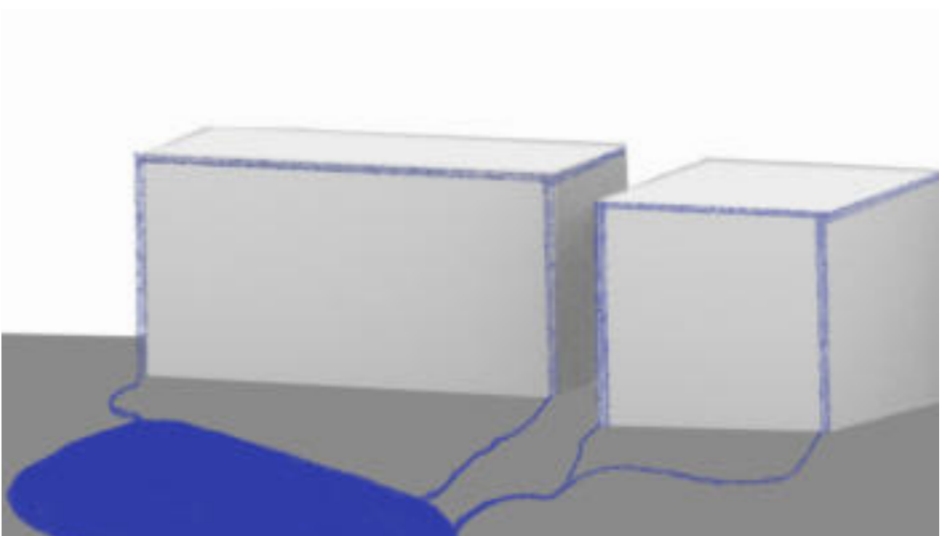
Separated Buildings

create better integration into a mixed-use neighbourhood



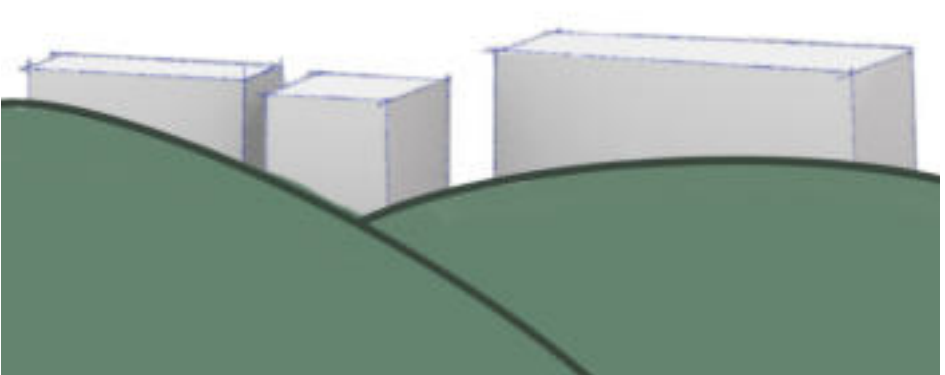
Green Areas

to improve biodiversity and integration into surrounding landscape



Rainwater collection pond

to improve local biodiversity and for emergency fires



Hill to hide buildings

to improve integration into landscape & hide buildings from street view

Traditional logistics facilities involve just one big mass that disrupts the view and blocks animals from passing through. Although separating the mass into smaller buildings is less efficient financially, it creates a more human scale facility, smoother integration into the surrounding area and allows small animals to pass through.

Green facades and roofs cool the building, while also providing shelter & food source for local birds and insects. Especially if native plants are being used, this naturally improves the local biodiversity. Furthermore, adding dedicated green zones throughout the facility promotes local biodiversity and creates a healthier work environment for staff.

Instead of letting rain water leak into the ground, it can be collected to form a pond nearby. This creates a calm environment for workers from the facility to relax at, and provides an excellent habitat for local frogs, insects and other animals.

local residents usually complain about logistics facilities being built nearby, because they obscure the view of the surrounding landscape. By excavating the soil where the main buildings will sit, and using it to create a small artificial hill in front of the buildings where they face residential areas, the sight of the buildings will be minimized.

Key Performance Indicators

energy consumption

space utilization

green space

carbon footprint

flexibility & expandability

hazard segregation

Green Facades

green facade trellis system

A green façade trellis system involves climbing plants growing from the ground or planting beds and attach to a secondary support structure, usually a steel cable or mesh grid fixed to the façade. The vegetation forms a living layer in front of the building envelope, creating a breathable buffer zone that improves environmental performance and softens the visual scale of large architectural buildings. This system is cheaper to operate than living wall panel systems and requires low maintenance.



Possible plants for the green facade

1 = very poor, 2 = weak, 3 = moderate, 4 = good, 5 = excellent



Celastrus Orbiculatus

Biodiversity Contribution	2/5
Thermal Performance	4/5
Growth & Coverage Efficiency	5/5
green facade Suitability	1/5



Aristolochia Macrophylla

Biodiversity Contribution	3/5
Thermal Performance	5/5
Growth & Coverage Efficiency	4/5
green facade Suitability	4/5



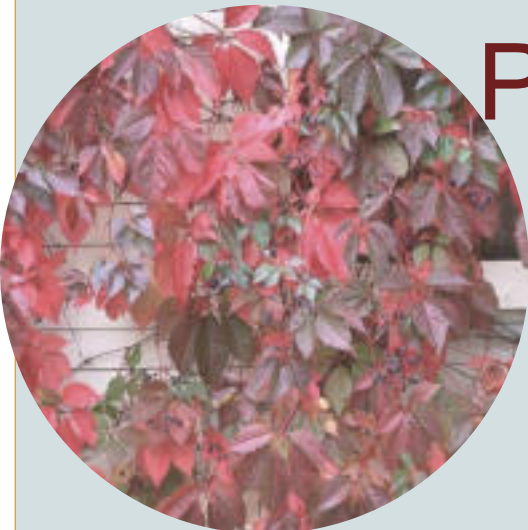
Fallopia Aubertii

Biodiversity Contribution	3/5
Thermal Performance	4/5
Growth & Coverage Efficiency	5/5
green facade Suitability	3/5



Hedera Helix

Biodiversity Contribution	5/5
Thermal Performance	4/5
Growth & Coverage Efficiency	3/5
green facade Suitability	5/5



Parthenocissus Quinquefolia

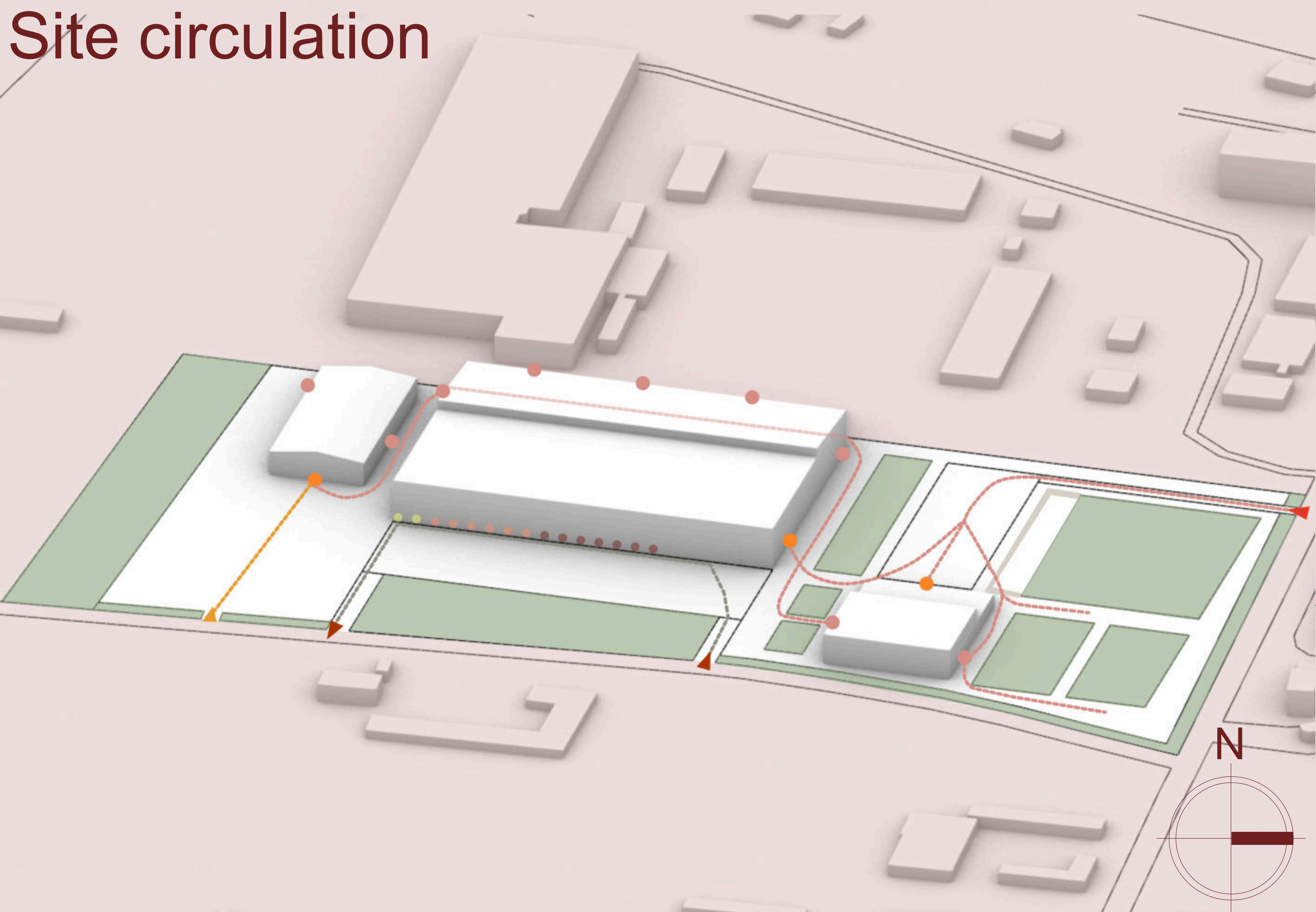
Biodiversity Contribution	4/5
Thermal Performance	4/5
Growth & Coverage Efficiency	5/5
green facade Suitability	5/5



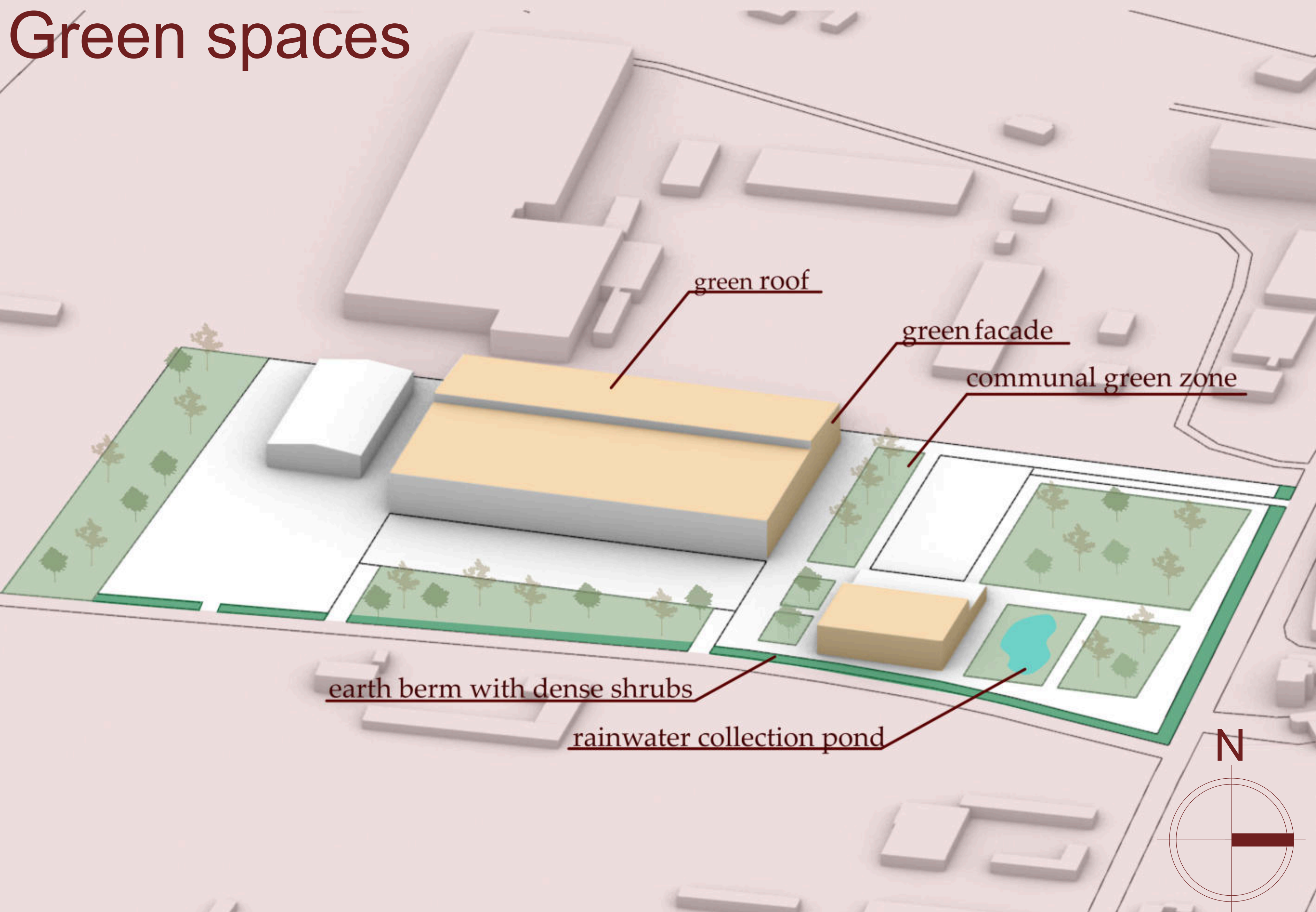
Lonicera Periclymenum

Biodiversity Contribution	5/5
Thermal Performance	3/5
Growth & Coverage Efficiency	4/5
green facade Suitability	5/5

Site circulation



Green spaces

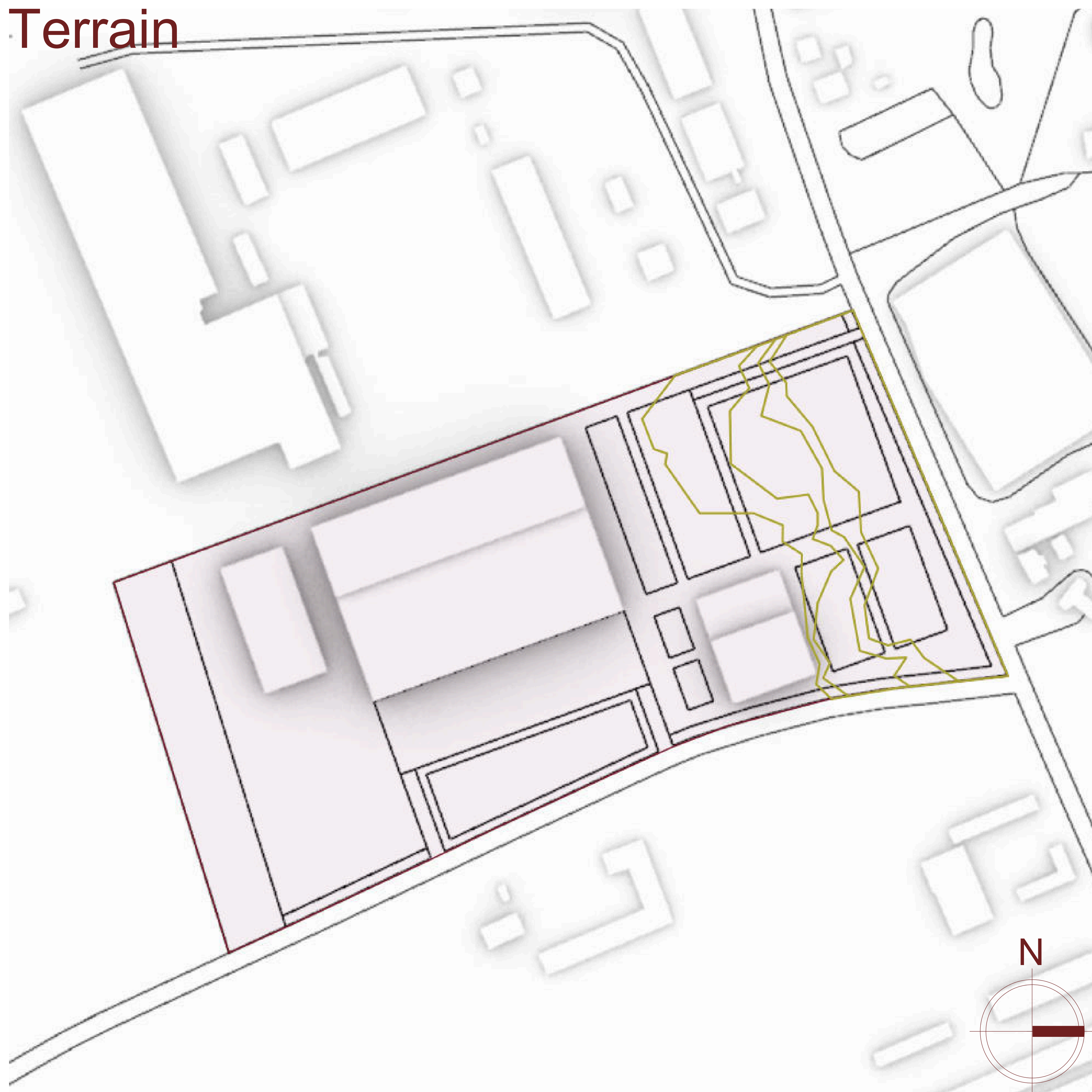


Legend

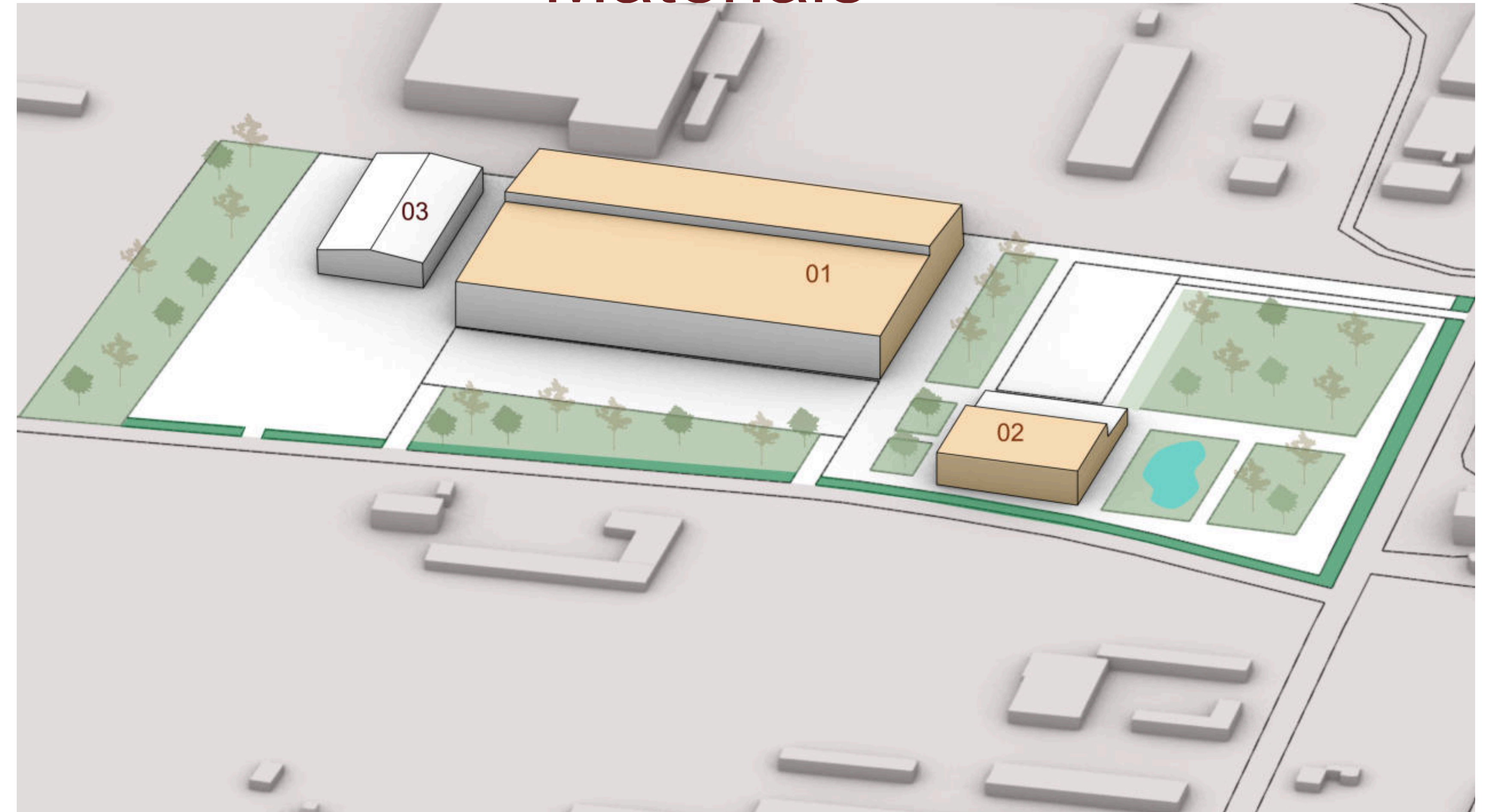
site entry	building entry	truck docks	circulation paths
▲ staff entry/exit	● main entrances	● inbound docks	● staff circulation
▲ Truck entry & exit	● secondary/emergency exits	● outbound docks	● truck circulation
▲ Fire truck entry/exit		● hazardous docks	● fire truck circulation

green roof	green facade	communal green zone	earth berm	rain water collection pond
The green roof on the processing building and office building will be covered in short shrubs and local wild flowers to serve as a food source for local insects & birds. The hazard storage building cannot have a green roof due to safety concerns.	The green facade covers all the side walls of the processing building except for the front side that has the loading docks as well as the sides of the office building will create passive cooling to the buildings, providing a comfortable environment for workers during the summer.	The communal garden will be accessible for workers from the recycling facility as well as local residents and will provide a calm place to relax away from the main logistics building. It will be planted with local tree & bush species to promote local wildlife.	The earth berm is 50cm tall and runs along the perimeter of the facility's north and east side that are facing towards public roads & residential housing. It serves as a visual barrier of the recycling facility. Together with the local bushes that will be planted on top, it will help reduce noise pollution from the recycling facility.	During heavy rain, water that falls on the office building will be diverted through gutters to the artificial pond adjacent to it. The pond will not only support the local ecosystem, but also serve as a water reserve for extinguishing potential fires in the hazardous storage facility.

Terrain



Materials



01 Main Processing Building

Structural columns: Structural steel (hot-rolled steel sections) – high load capacity and long spans for open floor plans

Walls: Precast reinforced concrete panels – impact-resistant and durable in industrial environments

Flooring: High-strength fiber-reinforced concrete with epoxy coating – resistant to abrasion, heavy loads, and chemical spills

Facade: Insulated metal sandwich panels – durable, thermally efficient, and easy to maintain

02 Admin Office Building

Structural columns: Reinforced concrete – stable, cost-effective, and fire-resistant

Walls (internal partitions): Gypsum board (drywall) – flexible for office layouts and easy to modify

Flooring: Raised access flooring (steel-core panels) – allows flexible cabling and IT infrastructure

Facade: Aluminium curtain wall with double glazing – maximizes daylight while maintaining thermal efficiency

03 Hazardous Waste Storage Building

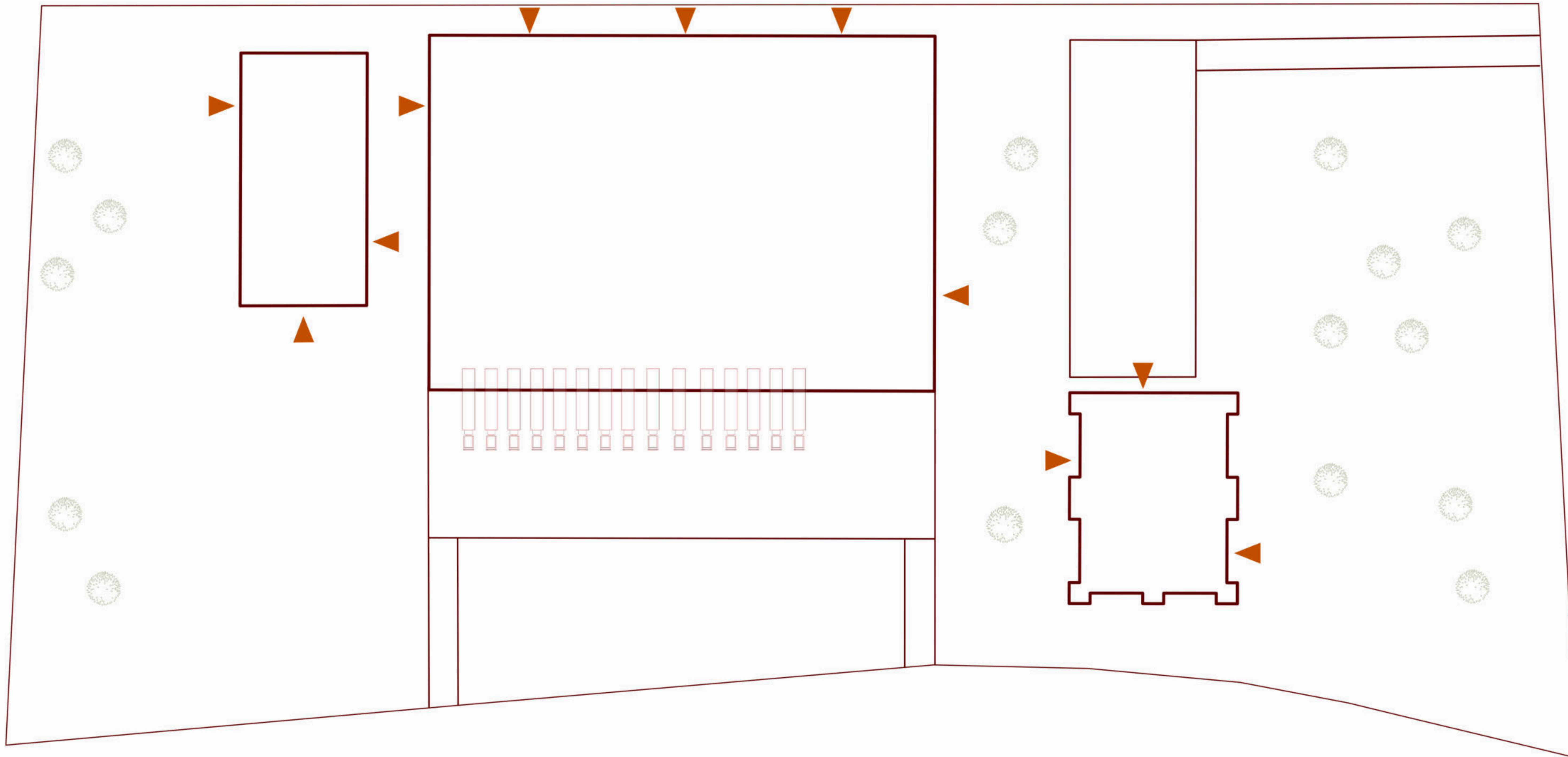
Structural columns: Reinforced concrete with chemical-resistant admixture – strong and resistant to corrosive exposure

Walls: Cast-in-place reinforced concrete with epoxy or polyurethane protective lining – ensures containment integrity

Flooring: Acid-resistant epoxy resin flooring with sealed joints – prevents leakage and resists chemical degradation

Facade: Precast concrete with protective chemical-resistant coating – robust external barrier with minimal permeability

Floor plan



site layout

the buildings on the site are strategically positioned so that the **hazardous waste building** is furthest away from the **office building**. This minimizes unnecessary worker exposure to chemical waste. The **hazardous waste building** is positioned strategically to allow firemen to quickly access it from the adjacent road in case of an emergency.

Building shapes

the **office building** features 6 nooks on floor 0 that create sheltered outdoor spaces, allowing workers to take breaks outside even during bad weather. It also features a roof terrasse with view onto the communal green zone, thereby creating a calm & comfortable environment for staff.

the **processing building** is split in two halves, with the front facing one having a lower total height. The back half has a higher total height, intended to house large machinery that would be unable to fit in the front half. Through this design, the building height is minimized where possible while still allowing for large machinery to fit into the facility.

Site access

There are three main entrances onto the site, strategically positioned to minimize traffic in the area. The staff entry is on the north of the facility, while the trucks enter on the east. The emergency fire entrance is also separate, located on the east.

Epilogue

The rapidly expanding EV market is increasing demand for battery recycling infrastructure, which presents a new architectural opportunity. As circular economies expand and material recovery becomes essential to future energy systems, recycling facilities can no longer remain spatially isolated and positioned at the margins of public and ecological life. Their continued development demands these facilities to be rethought in a way that balances industrial performance with environmental and residential integration.

This project redevelops the battery processing facility as an active component of the urban and ecological landscape rather than a purely operational infrastructure. Located within the transitional industrial-residential edge condition of Česká Lípa, the project proposes topographic embedding, fragmented massing, green façade & roof systems, and a rainwater collection pond. By partially excavating the site and constructing protective green berms, the industrial scale of the facility is visually reduced and integrated into the surrounding terrain, allowing logistics architecture to operate with greater reciprocity toward its context.

The separation of administrative, processing, and hazardous storage functions into distinct but connected volumes introduces a more human-scaled spatial hierarchy, improving daylight access, worker comfort, and environmental legibility. Green façade trellis systems and green roofs reduce thermal gain, supporting biodiversity, and softening the visual presence of technical infrastructure.

Ultimately, the project argues that logistics architecture should be seen as a civic and environmental system capable of coexistence with surrounding communities and landscapes. The project proposes an alternative model for logistics development, where technical performance, ecological regeneration, and human experience are able to work together rather than be seen as conflicting objectives.

